

Pennsylvania Embraces Home Energy

Rising energy prices and environmental concerns have increased interest among consumers, design professionals, and the trades in green building technology and energy efficiency—or green energy, for short. In response to this combination of need and demand, the West Penn Power Sustainable Energy Fund (WPPSEF) has developed PA Home Energy, a new program to help link consumers to energy consultants and contractors who can provide high-quality green energy services.

PA Home Energy uses a comprehensive whole-house approach that analyzes the overall performance of the home rather than simply evaluating a single component, such as a furnace. The program is unique in that it targets both existing and new homes in order to provide a comprehensive program that reaches all Pennsylvanians. PA Home Energy is supported by the U.S. Environmental Protection Agency (EPA) and the U.S. Department of Energy (DOE). It makes extensive use of the experience and materials already developed for these agencies' Energy Star Qualified New Homes program and their related program for existing homes, Home Performance with Energy Star. Other partners in the development of PA Home Energy include Allegheny Power, Affordable Comfort, Incorporated (ACI), and AFC First Financial.

PA Home Energy is being piloted in the 23 counties where Allegheny Power provides electric service to over 710,000 residential customers. Although the program is being piloted on a regional basis, program partners hope to obtain additional resources that will enable them to expand the program statewide.

Developing Knowledgeable Contractors

One of the hallmarks of both of the Energy Star programs is the requirement that all homes meet stringent energy efficiency standards. To ensure that consultants and contractors know what needs to be done to meet these standards, the WPPSEF is providing incentives to Pennsylvania-based companies to train their employees in building science and home performance techniques. To prove competency, PA Home Energy is requiring its registered consultants and contractors to obtain certification from two certifying organizations, the Building Performance Institute (BPI) and the Residential Energy Service Network (RESNET) home energy rating organization.

Participating consultants and contractors are trained to look at whole-house performance. They learn to use building science and green energy technology to prevent and solve a wide range of problems, including ice dams, drafts, poor indoor air quality (IAQ), mold, dust, and risk of carbon monoxide (CO) exposure—all while saving significant energy dollars. They learn how to use modeling software to calculate the energy savings from a wide range of energy improvements. They also learn how to use the software to evaluate the cost-effectiveness of various renewable energy technologies, such as geothermal heat pumps and solar photovoltaic systems.

When they are called to an existing home, participating contractors have been required since November 2007 to conduct a green energy audit. This audit provides homeowners with an unbiased assessment of their current



Brian Henderson of State College-based Envinity places the Energy Star Qualified Home sticker on the first Energy Star Qualified Home in the PA Home Energy program.

energy use, and lists the steps that they can take to improve home performance and lower their utility bills. PA Home Energy green energy audits help homeowners to prioritize their energy improvements, based on their budget and their individual needs. These improvements might include air sealing the house envelope, adding insulation, installing high-efficiency heating and cooling systems, replacing windows and doors, or installing renewable energy technologies. Participating contractors also routinely compare the costs of, and the energy savings to be realized by, installing each of these various improvements.

Consumer-Based Incentives

The WPPSEF provides modest consumer-based incentives to homeowners in the pilot program to encourage them to install energy improvements. The incentives range from \$200 to \$1,000, based on how much energy the improvements are expected to save. Energy savings are determined by PA Home Energy registered personnel, using the modeling software described above. An Energy Savings Certificate is issued to the owners of existing homes, as a way to reinforce the importance of making these improvements. PA Home Energy

also helps homeowners to register their newly constructed homes to obtain an Energy Star label.

Market Drivers

The market for Energy Star-qualified new homes is growing rapidly. Reduced demand for new homes, combined with rising energy costs and concern about global warming, is increasing the need for builders to differentiate the energy and environmental performance of the homes they build. Builders who participate in the Energy Star Qualified New Homes program get this marketing advantage and also benefit from reduced callbacks. For both of these reasons, Energy Star Qualified New Homes achieved market penetrations as high as 60% in some parts of the country and 12% nationally, in 2006. In Pennsylvania, the state housing finance agency has recognized the need to control the energy costs of affordable housing and has moved to require that homes financed through the agency be Energy Star qualified.

The growth of the Home Performance with Energy Star program is likewise based on market drivers that are affecting the home improvement industry. These market drivers include high energy prices, customer demand for comfort and improved IAQ, and more investment in remodeling as land and housing prices spiral upward. The whole-house approach bundles all these consumer interests into a high-value package that offers customers a one-stop-shop opportunity. The use of performance testing and the consultant's or contractor's knowledge of building science builds customers' trust and makes it easier to convince them to move forward with installations. Home Performance with Energy Star further develops consumer trust by requiring third-party inspection of completed jobs for performance and safety. The PA Home Energy program gives participating contractors access to all the advantages of the Home Performance with Energy Star program.

Participating in PA Home Energy enables contractors to respond to these market pressures and grow their business in an expanding green energy market, while increasing their understanding of how buildings work. More importantly, it enables them to help Pennsylvania homeowners lower their energy bills by reducing their energy consumption and the related emissions of greenhouse gases. With this program, all Pennsylvanians win.

H_e

—Joel L. Morrison

Joel L. Morrison is a research associate at the Pennsylvania State University's Energy Institute and is the director of the West Penn Power Sustainable Energy Fund.

For more information:

To learn more about PA Home Energy, go to www.pahomeenergy.org.